

SALAD: SOC Alert Labeled Analysis Dataset — A Unified Benchmark for Alert Triage, Prioritization, and Classification

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Abstract

This paper presents SALAD (SOC Alert Labeled Analysis Dataset), a unified benchmark dataset containing 2,778,424 SOC (Security Operations Center) alerts derived from two established network intrusion detection datasets: CIC-IDS2017 and UNSW-NB15. Each alert record follows a standardized 29-field schema and is enriched with MITRE ATT&CK tactic and technique mapping, Cyber Kill Chain phase labels, five-level severity scoring, analyst triage decisions (escalate, investigate, or suppress), continuous priority scores, three-level difficulty labels, and natural language alert descriptions. The dataset is split into training (70%), validation (15%), and test (15%) sets with stratified sampling. Four benchmark tasks are defined: binary alert classification, three-class triage prediction, priority score regression, and 15-class attack categorization. Base-line evaluation scripts using Logistic Regression and Random Forest classifiers are provided. The dataset, evaluation code, and data card are publicly available on Zenodo (DOI: 10.5281/zenodo.18912306), HuggingFace, and GitHub under a CC-BY 4.0 license.

Keywords: SOC alert triage, MITRE ATT&CK mapping, cybersecurity benchmark, network intrusion detection, security operations center, multi-task evaluation

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1. Specifications Table

Subject	Computer Science / Cybersecurity
Specific subject area	SOC alert triage, intrusion detection, cybersecurity benchmark
Type of data	CSV (tabular), JSON (statistics)
Data collection	Derived from CIC-IDS2017 and UNSW-NB15 via an automated five-stage conversion pipeline that adds SOC-specific metadata (triage decisions, priority scores, MITRE ATT&CK mapping, difficulty levels, and natural language descriptions) not present in the source data
Data source location	CIC-IDS2017: Canadian Institute for Cybersecurity, University of New Brunswick, Canada; UNSW-NB15: University of New South Wales, Australia
Data accessibility	Repository name: Zenodo, HuggingFace, GitHub DOI: 10.5281/zenodo.18912306 Zenodo: https://doi.org/10.5281/zenodo.18912306 HuggingFace: https://huggingface.co/datasets/nutthakorn7/SALAD-SOC GitHub: https://github.com/nutthakorn7/SALAD-SOC
Related research article	N/A (standalone data article)

2. Background

Security Operations Centers (SOCs) process thousands to millions of alerts daily from intrusion detection systems, firewalls, and endpoint security tools [1]. Studies report that SOC analysts face an average of 4,484 alerts per day, with 67% of alerts being ignored due to alert fatigue [2]. Efficient alert triage — deciding which alerts to escalate, investigate, or suppress — is critical for timely incident response [3]. Although ML-based [4] and RL-based approaches [5] to alert triage automation are gaining traction, most studies rely on proprietary data or synthetic simulations because no public SOC alert dataset with triage

labels exists [6, 7]. Existing public datasets such as CIC-IDS2017 [8] and UNSW-NB15 [9] provide raw network traffic but lack SOC-specific metadata: there are no triage decisions, no priority scores, no MITRE ATT&CK mapping [10], and no structured benchmark tasks. SALAD fills this void by transforming these raw traffic records into a realistic SOC alert format, adding five layers of computed metadata that do not exist in the source data.

3. Value of the Data

- SALAD provides real network traffic transformed into a unified SOC alert format combining MITRE ATT&CK mapping, triage decision labels, priority scores, difficulty levels, and natural language descriptions — properties not simultaneously available in existing public datasets (see Table 4).
- The dataset enables researchers across multiple communities — machine learning, reinforcement learning (RL/MARL), and large language models (LLMs) — to develop and benchmark alert triage systems using a single, standardized evaluation framework [11].
- With 2,778,424 alerts across 15 attack categories from two established source datasets, SALAD provides sufficient scale and diversity for training deep learning models while maintaining reproducibility through deterministic generation from public source data.
- Difficulty levels (easy/medium/hard) and LLM-ready natural language descriptions open up new evaluation setups, such as difficulty-stratified benchmarking and prompt-based alert triage.
- SALAD provides a reproducible public benchmark for RL/MARL-based alert triage research, where existing datasets are either proprietary, synthetic, or lack the multi-task evaluation structure needed for systematic comparison.

4. Data Description

4.1. Dataset Overview

SALAD contains 2,778,424 SOC alerts derived from two established network intrusion detection datasets: CIC-IDS2017 [8] (2,520,751 alerts) and UNSW-NB15 [9] (257,673 alerts). Table 1 summarizes the dataset.

Table 1: SALAD Dataset Overview

Metric	Value
Total alerts	2,778,424
Columns (fields)	29
Attack categories	15
MITRE ATT&CK tactics	9
MITRE ATT&CK techniques	12
Kill chain phases	8
Malicious ratio	21.2%
Benign ratio	78.8%
Attack campaigns (correlated)	586,227
File size (full dataset)	878 MB
Training split	1,944,887 (70%)
Validation split	416,756 (15%)
Test split	416,781 (15%)

4.2. Schema Design

Each alert record contains 29 fields organized into six groups (Table 2).

4.3. Benchmark Tasks

SALAD defines four evaluation tasks (Table 3).

4.4. Comparison with Existing Datasets

Table 4 compares SALAD with nine existing cybersecurity datasets [13, 8, 9, 7, 6, 3, 14, 15, 16, 5] across 10 key dimensions.

5. Experimental Design, Materials, and Methods

5.1. Source Datasets

Two widely-cited, publicly available network intrusion detection datasets are used as source data:

1. **CIC-IDS2017** [8]: Contains network traffic captured over five days (July 3–7, 2017) at the Canadian Institute for Cybersecurity. Includes 2,520,751 labeled flows across 14 attack types (grouped into 7 categories in the original paper).
2. **UNSW-NB15** [9]: Contains 257,673 records combining normal and attack traffic across 9 attack categories, generated using the IXIA PerfectStorm tool at the University of New South Wales.

5.2. Conversion Pipeline

The SALAD generation pipeline consists of five stages:

Stage 1 — Schema Mapping: Each source record is mapped to the SALAD 29-field schema. Attack labels are mapped to standardized `alert_type` categories, MITRE ATT&CK tactics and techniques, and Cyber Kill Chain phases. IP addresses are anonymized using deterministic hashing for reproducibility. Severity levels are assigned based on attack type impact classification.

Stage 2 — Confidence Scoring: Detection confidence scores are computed as: $c = \text{base}(m) + 0.3 \cdot s + 0.2 \cdot \min(1, f/10^6) + \epsilon$, where m indicates malicious status (base = 0.3 for malicious, 0.15 for benign), s is the severity score, f is the flow bytes per second, and $\epsilon \sim U(-0.05, 0.05)$ adds realistic noise.

Stage 3 — Triage and Priority: Triage decisions (escalate, investigate, suppress) are assigned using rule-based policies derived from SOC triage guidelines. Priority scores are computed as: $p = 0.4s + 0.3c + 0.2m + 0.1 \min(1, n_{1h}/10)$, where n_{1h} is the temporal alert count. Difficulty levels are assigned based on

the combination of severity, confidence, and malicious status, producing easy (60.1%), medium (33.0%), and hard (6.9%) distributions.

Stage 4 — Enrichment: Temporal context features (`alert_count_1h`, `alert_count_24h`) are computed via per-source-IP rolling windows. Attack campaigns are identified using correlation keys based on source IP and kill chain phase combinations, yielding 586,227 unique campaigns. Natural language alert descriptions are generated from templates specific to each alert type.

Stage 5 — Splitting: Stratified train (70%) / validation (15%) / test (15%) splits are generated using joint stratification [17] by `attack_category` and `source_dataset` to ensure consistent class distributions across splits.

Label Validation

The rule-based triage labels were validated through three checks: (1) all escalate-labeled alerts correspond to high- or critical-severity attack types, matching standard SOC triage protocols; (2) the class distribution (escalate 14.4%, investigate 6.9%, suppress 78.8%) is consistent with reported real-world SOC alert distributions where 60–90% of alerts are false positives or low-priority [1, 2]; and (3) the baseline Random Forest classifier achieves a weighted F1 of 0.9751 on triage prediction (Table 5), confirming that the labels encode learnable patterns from network-level features rather than random assignments.

5.3. Baseline Evaluation

We evaluate Random Forest [18] (100 trees) on all four tasks using only network-level features (ports, flow statistics, temporal counts) to avoid data leakage from label-derived features [19]. Table 5 presents the detailed results.

5.4. Reproducibility

The complete pipeline is implemented in Python and publicly available at <https://github.com/nutthakorn7/SALAD-SOC>. Running `build_salad_fast.py` with the two source datasets (both publicly downloadable) deterministically reproduces the identical SALAD dataset. The build process completes in approximately 266 seconds on a standard workstation (Intel i7, 16 GB RAM). Baseline

evaluation can be reproduced using `evaluate.py --task all --max-train 50000`.

5.5. File Structure

The released dataset contains the following files:

- `salad_train.csv` — Training split (1,944,887 alerts, 615 MB)
- `salad_val.csv` — Validation split (416,756 alerts, 132 MB)
- `salad_test.csv` — Test split (416,781 alerts, 132 MB)
- `stats.json` — Dataset statistics summary (JSON)
- `datacard.md` — Detailed dataset documentation
- `evaluate.py` — Evaluation and baseline scripts (Python)
- `LICENSE` — CC-BY 4.0 license

6. Ethics Statement

SALAD is derived entirely from two widely-used, publicly available datasets originally created for network security research (CIC-IDS2017 and UNSW-NB15). No new data collection involving human participants or animals was performed. All IP addresses in the source datasets are anonymized using deterministic hashing. No personally identifiable information (PII) is present in the released dataset. As a secondary analysis of existing public datasets, this work did not require institutional ethics review board approval. The dataset is released under a CC-BY 4.0 license.

7. Limitations

The dataset has several limitations that users should consider. First, SALAD alerts are synthetically derived from CIC-IDS2017 and UNSW-NB15 network

traffic rather than captured from a live SOC environment, so the alert distribution and temporal patterns may not fully reflect production SOC workloads [20]. Second, the triage decisions (escalate/investigate/suppress), severity scores, and priority labels are generated by deterministic rules based on attack type rather than labeled by human SOC analysts, which may introduce systematic biases absent from real analyst judgments. Third, the natural language alert descriptions are template-generated and lack the variability of vendor-specific alert formats. Finally, the dataset does not include contextual information such as asset criticality, network topology, or historical incident data that analysts typically use in triage decisions.

8. CRediT Author Statement

Nutthakorn Chalaemwongwan: Conceptualization, Methodology, Software, Data Curation, Validation, Writing – Original Draft, Writing – Review & Editing.

9. Declaration of Competing Interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 2: SALAD Schema (29 fields)

Group	Field	Description
Metadata	alert_id	Unique identifier
	timestamp	ISO 8601 timestamp
	source_dataset	CICIDS2017 or UNSW_NB15
Network	src_ip, dst_ip	Anonymized IP addresses
	src_port, dst_port	Port numbers
	protocol	TCP, UDP, etc.
	flow_duration	Duration in seconds
	total_fwd/bwd_packets	Packet counts
	flow_bytes_per_sec	Throughput
Alert	alert_type	16 alert categories
	severity	none/low/medium/high/critical
	confidence	Detection confidence [0,1]
	mitre_tactic/technique	MITRE ATT&CK v14 mapping [10]
	kill_chain_phase	Lockheed Martin kill chain + unknown [12]
Context	network_segment	dmz/server/workstation/mgmt
	alert_count_1h/24h	Temporal alert frequency
	is_repeated_target	Repeated target indicator
	correlation_key	Attack campaign identifier
Labels	is_malicious	Ground truth binary label
	attack_category	15 attack categories
	triage_decision	escalate/investigate/suppress
	priority_score	Weighted priority [0,1]
LLM	difficulty_level	easy/medium/hard
	alert_description	Natural language description

Table 3: SALAD Benchmark Tasks and Baseline Results

Task	Type	Target	Metric	RF Baseline
1. Alert Classification	Binary	is_malicious	F1	0.9580
2. Alert Triage	3-class	triage_decision	Weighted F1	0.9751
3. Prioritization	Regression	priority_score	Spearman	0.7617
4. Attack Category	Multi-class	attack_category	Macro F1	0.5479

Table 4: Comparison with Existing Datasets ($\checkmark$ = fully supported, $\sim$ = partial, $\times$ = not available)

Dimension	KDD	CIC-IDS	UNSW	MS GUIDE	AIT-ADS	SABU	UWF-Zeek	Mordor	Cyborg	SALAD
Public download	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Real traffic base	$\checkmark$	$\checkmark$	$\sim$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\checkmark$
SOC alert format	$\times$	$\times$	$\times$	$\sim$	$\sim$	$\sim$	$\times$	$\times$	$\times$	$\checkmark$
Triage labels	$\times$	$\times$	$\times$	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
MITRE ATT&CK	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$	$\sim$	$\checkmark$
Unified schema	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	N/A	$\checkmark$
Multi-task benchmark	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
RL/MARL ready	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
LLM eval-ready	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Reproducible	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\sim$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Total $\checkmark$	4	4	4	4	2	2	5	3	4	10

Table 5: Detailed Baseline Results — Random Forest (100 trees), test set

Task	Class/Metric	Precision	Recall	F1	Support
1. Classification	Benign	0.98	0.99	0.99	328,211
	Malicious	0.97	0.93	0.95	88,570
	<i>Weighted avg</i>	<i>0.96</i>	<i>0.96</i>	<i>0.96</i>	416,781
2. Triage	Escalate	0.95	0.94	0.94	59,993
	Investigate	0.89	0.88	0.89	28,577
	Suppress	0.99	0.99	0.99	328,211
	<i>Weighted avg</i>	<i>0.98</i>	<i>0.98</i>	<i>0.98</i>	416,781
3. Prioritization	Spearman ρ	0.7617			
4. Attack Category	Macro F1	0.5479 (15 classes)			